

CSC494

Dirt Bike Maintenance Sensor System

Challenges

Integrating sensors into a real dirt bike revealed challenges not apparent during initial design.

- The core problem — avoiding invasive modifications:
- Preserving the structural integrity of engine components
- Preventing potential leaks or contamination
- Maintaining warranty and resale value
- Minimizing risk of mechanical failure during aggressive off-road use

Alterations in Plans

Original design included:

- Ultrasonic oil level sensor inside the oil pan
Coolant level sensor in the radiator/overflow reservoir

Why these were dropped:

- Oil level sensor requires access inside the oil pan — a custom oil fill cap was considered but requires fabrication tools not currently available
- Coolant sensor has the same constraints — no way to access the reservoir without drilling or permanent modification

These mounting challenges are the primary reason the project scope was changed.

New Plan

Refocused on two sensors that can be mounted right now:

- MPXV7002DP air pressure sensor — two small silicone tubes, zero permanent modification to the bike
- DS18B20 temperature probe — clamped or zip-tied to the outside of the oil pan to read surface temperature

Prototype goal:

- Prove the concept works with the bike stationary — show that pressure changes measurably when the air filter is restricted or dirtied, and that temperature tracks engine warm-up correctly. This validates the core sensing logic before tackling permanent mounting.

Progress

- Connected MPXV7002DP differential pressure sensor to Seeed Studio XIAO ESP32-C6
- Successfully reading live air box pressure with automatic zero calibration at boot
- Connected DS18B20 waterproof temperature probe via Arduino Nano to XIAO over serial
- Nano reads oil temperature and transmits to XIAO every second
- Live JSON telemetry payload transmitting pressure and temperature every 500ms
- Threshold alerts implemented for filter status (CLEAN / WATCH / WARNING / CRITICAL) and oil temperature (COLD / NORMAL / WARNING / CRITICAL)
- Wiring was harder than expected — voltage levels, pull-up resistors, and pin compatibility required significant troubleshooting

Known Unknowns

- Building a mobile dashboard to display live BLE telemetry visually
- Mounting the sensors on an actual dirt bike in a waterproof enclosure
- Power source independent of a laptop — bike battery or small dedicated battery
- Validating pressure thresholds against a real dirty vs clean air filter

Unknown Unknowns

- Whether MPXV7002DP readings will remain stable under engine vibration
- How heat from the engine bay will affect sensor accuracy over time
- How accurate and generalizable the readings will be across different bike models

Learning with AI

Making stuff up:

- AI sometimes suggests incorrect pin numbers or library functions that do not exist

Hardware compatibility gaps:

- DS18B20 could not be read directly from the XIAO — required offloading to the Arduino Nano

What AI was good at:

- Diagnosing serial monitor output and identifying root causes quickly
- Rewriting specific functions without breaking the rest of the code
- Explaining wiring step by step with clear tables

Sprint 2

- Build a mobile BLE dashboard to display live - pressure and temperature readings
- Test the system on a real dirt bike engine
- Investigate waterproof enclosure options for field deployment
- Validate pressure thresholds using a real dirty air filter